CS^{n}Net: A Remote Sensing Detection Network Breaking the Second-Order Limitation of Transformers With Recursive Convolutions

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***Abstract*— In recent years, Transformer-based networks, known for their ability to model long-range dependencies, have been widely used in downstream computer vision tasks, surpassing certain neural network architectures. However, Transformer-based networks suffer from issues, such as large parameter size, high-computational complexity (CC), and difficulties in extending spatial and channel features to the third or even higher orders, resulting in convergence challenges for small- to medium-sized datasets and limited effectiveness in extracting high-order detailed features. In this article, we propose a high-order spatial and channel controllable convolution module, named CS^n, which can replace standard convolutions in any convolutional network. In the context of remote sensing small object detection, CS^n demonstrates superior performance compared with self-attention structures embedded in neural networks. Moreover, it introduces long-range dependency relationships among pixels, similar to self-attention, and adopts a cascaded recursive approach to extend spatial and channel features to arbitrary higher orders without introducing significant additional computation. This extension captures crucial information from high-order spatial and channel dimensions, resulting in improved accuracy for small object detection. In addition, we construct a novel, versatile CS^n-[feature pyramid network (FPN) + path aggregation network (PAN)] structure for object detection networks, referred to as CS^{n}Net. Finally, our proposed model exhibits significant advantages in remote sensing detection when compared with state-of-the-art methods on publicly available synthetic aperture radar (SAR) datasets [SAR ship detection dataset (SSDD) and high resolution SAR images dataset (HRSID)] and optical remote sensing dataset [Northwestern Polytechnical University very-high-resolution-10 dataset (NWPU-10)], achieving respective improvements of 3.4%, 4.5%, and 0.4% in mAP$_{50}$ compared with baseline models.**

***Index Terms*— Convolutional neural networks (CNNs), feature aggregation, high-order feature interaction, remote sensing target detection, Transformer architecture.**

NOMENCLATURE

Evaluation index	Meaning
AP	Average precision.
mAP	Mean average precision.
Paras	Number of parameters for the network.
GFLOPs	Giga floating-point operations per second.
FPS	Frames per second.

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I. INTRODUCTION

WITH the development of globalization in recent years, maritime trade has exerted a profound effect on global economic development. According to data from the United Nations Conference on Trade and Development, maritime trade accounts for about 80% of global trade. Furthermore, the issue of maritime vessel traffic regulation arising from oceanic trade has become increasingly prominent.

Various methods are currently available for maritime vessel regulation, including visual monitoring [1], [2], satellite remote sensing [3], [4], and automatic identification system (AIS) monitoring. Managers monitor the location and trajectory of vessels to ensure that they are traveling on designated routes and to identify potential safety hazards. Although visual monitoring technology is easy to deploy and develop, it is susceptible to weather conditions. The AIS monitoring system has a wide monitoring range, high real-time performance, and low cost, but it can only monitor vessels equipped with AIS devices and is also vulnerable to data tampering. A synthetic aperture radar (SAR) is a method of remote sensing using radar technology for long-range imaging. Unlike optical remote sensing technology, SAR can acquire remote sensing data independently of lighting conditions, weather, and cloud cover, and can obtain high-resolution ground information in any weather, thus providing unique advantages.

At present, there are detection methods for remote sensing (optical and SAR) images based on traditional features [5], [6], [7], [8] and deep learning [9], [10], [11], [12]. Traditional methods use manually extracted image features (such as shape, texture, and edges) to recognize targets. After extracting features using methods, such as wavelet transform, classifiers, such as support vector machines (SVMs) and artificial neural networks (ANNs), are used for classification and recognition. Therefore, traditional methods have the shortcomings of requiring artificial design of features, high data quality requirements, and poor model scalability, which cannot fully exploit the potential of data and improve the performance of classifiers. In recent years, deep learning has made great progress in the field of object detection, improving the accuracy of detection, but this method still has problems, such as

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missing small targets, false detection, and poor detection real-time performance.

Small object detection plays a crucial role in remote sensing imagery due to the presence of numerous small-sized objects, such as ships, vehicles, buildings, and farmlands. Accurately detecting these small objects holds significant research significance in various fields, including maritime security monitoring, urban planning, traffic surveillance, and agricultural management. However, small targets, such as ships, have the characteristics of multiscale and biased distribution toward small targets, which exacerbates the above problems in the remote sensing field. In order to further solve the above problems, domestic and foreign researchers are committed to strengthening the feature extraction of images, enhancing the network's ability to learn features, and solving problems, such as small target recognition. For example, the performance of small target detection is improved by adopting deeper convolutional neural networks (CNNs) structures [13], [14], [15], increasing attention mechanisms [16], [17], [18], [19], adding skip connections [20], [21], optimizing convolutional modules [22], [23], and other methods to further extract advanced semantic information; or by using Transformer architectures with strong parallelization capabilities that can capture global contextual information and are advantageous for handling long-distance dependency problems to improve the performance of small target detection.

At present, many experiments have shown that the Transformer structure [24], [25], [26] has outperformed CNNs in tasks, such as object detection. This is mainly attributed to its second-order self-attention mechanism, which can capture the interaction of second-order spatial features, and their large convolutional kernel [27], which can provide a larger receptive field and obtain better contextual information. Subsequent researchers have proposed hybrid models that efficiently combine the advantages of Transformer and CNNs. However, The utilization of a 1-D sequence in Transformer models for image processing leads to the unfortunate loss of essential 2-D structural information. While CNNs excel at harnessing the inherent 2-D structure within images, Transformer models, relying on a 1-D sequence representation, struggle to effectively capture the spatial relationships crucial to understanding the content of the images. Furthermore, the CC of Transformer models is quadratic, which makes it expensive and challenging to train when dealing with high-resolution images. As the image size increases, the computational cost significantly rises, demanding more computational resources. In addition, compared with traditional CNNs, Transformer models typically have a larger number of parameters. In addition, while Transformer models consider second-order spatial adaptivity, they overlook channel adaptivity as well as higher order adaptivity between spatial and channel dimensions. This limitation becomes particularly pronounced when processing extremely long sequences, increasing the model's demand for computational resources. As a result, the quadratic complexity of the computations becomes prohibitively expensive, restricting the scalability of the model. This constraint hinders the model's ability to fully capture the higher order interaction information between channels and spatial dimensions.

In this article, we combine the strengths of two models and make the following contributions.

1) We design a high-order neural network module, called CS^n (arbitrarily high-order spatial and channel interaction convolution module), which is inspired by the Transformer concept. It enables feature interactions in arbitrary high-order spatial and channel dimensions, achieving comparable feature extraction performance to the self-attention module in Transformer networks without significantly increasing the network parameters. Building upon this, we further devise CS^nNet, which incorporates the CS^n-[feature pyramid network (FPN) + path aggregation network (PAN)] structure, allowing us to extract valuable information from high-order spatial and channel dimensions.
2) We propose a novel method for decomposing large convolution kernels, treating them as a combination of depthwise separable convolution, dilated convolution, and pointwise convolution. To avoid feature misalignment, we introduce a group convolution and feature fusion approach.
3) We demonstrate that our method achieves the state-of-the-art performance on publicly available SAR datasets, such as SAR Ship Detection Dataset (SSDD) and high resolution SAR images dataset (HRSID), as well as optical remote sensing dataset Northwestern Polytechnical University very-high-resolution-10 dataset (NWPU-10).

II. Related Work

Small object detection in remote sensing has significant application value in many fields, such as military, security, environmental monitoring, resource management, disaster monitoring, urban intelligence, and traffic management. It plays a crucial role in safeguarding national security, promoting national progress, and improving people's quality of life. However, there is still considerable room for improvement in remote sensing small object detection. The unsatisfactory performance can mainly be attributed to two aspects. First, small targets are frequently multiscale and occupy only a small number of pixels in an image. As the resolution of feature maps decreases, target information is easily lost. Second, the limited receptive field caused by small convolutional kernels in convolutional networks makes missing global information easy, resulting in the detection network being susceptible to noise from complex backgrounds and leading to decreased detection performance.

In contrast with convolutional operations that only model local dependencies, self-attention models global pairwise dependencies using large patches. It provides a more effective method for encoding 2-D spatial information, allowing features to interact in 2-D space. Due to the nonlinearity of features, high-order spatial features often contain information with varying degrees of correlation. This is why self-attention can demonstrate advantages. However, a major drawback of self-attention is its high-computational cost. So, how can we strike a balance between computational resources and performance? To address this issue, we summarize the methods for small object detection into three categories in this article.

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A. Models Based on CNN Structure

Target detection models based on convolutional structures can be divided into two main types. The first type is the convolutional models that utilize CNNs for the task of object detection. These models are typically categorized into two-stage and one-stage detection models. Two-stage detection models divide the object detection task into two subtasks: generating a set of candidate bounding boxes and classifying/regressing these boxes. Examples include Fast region based CNN (R-CNN) [28], Faster R-CNN [11], and Mask-CNN [29]. These models often achieve higher localization accuracy but require more computational resources and time. Meanwhile, one-stage detection models can quickly detect all objects and perform well in dense object scenarios, but they generally have lower localization accuracy. Examples of one-stage detection models include single shot detector (SSD) [13], RetinaNet [12], and you only look once (YOLO) series [14], [15], [30], [31], [32], [33], [34].

In general, CNNs enable the extraction of features with translation and rotation invariance, contributing significantly to the field of object detection. However, with the growing contributions of Transformer-based networks [24] in computer vision, traditional convolutional networks have also gained attention from researchers. This situation has led to the emergence of the second type of convolutional models, known as Transformer-style convolutional models. For example, ConvNext [35] incorporates the latest ideas and techniques from Transformer networks into the existing CNN modules, combining the advantages of both networks and improving the performance of CNNs. Conv2Former [36] aims to simplify self-attention by modulating the V matrix in the module with convolutional features, reducing the model parameters while maintaining accuracy. HorNet [37], from the perspective of spatial interaction modeling, extends the second-order spatial interaction modeling of self-attention to higher orders using convolutional operations without significantly increasing network computation, thereby enhancing the detection performance of the network.

These successful experiments demonstrate the usefulness of transferring the advantages of self-attention to CNNs through convolutional operations, inspiring further research in this direction.

B. Models Based on Transformer Structure

Most of the models used in object detection tasks are based on CNNs, while the Transformer model [38], [39], [40] is mainly used in natural language processing (NLP) tasks. However, the self-attention mechanism in the Transformer architecture has been widely applied to computer vision, bringing significant improvements to image processing and visual reasoning tasks. For example, the vision Transformer (ViT) [41] divides the image into multiple patch sequences, achieving a conversion from 2-D to 1-D sequences, which enables the image to obtain long-distance dependencies, but the resulting CC is $O(N^2)$. Swin Transformer [42] proposed a hierarchical network structure to address the multiscale problem of visual images, resulting in outstanding performance in object detection tasks, but it did not consider the interrelationships between patches and did not extract target features for different scale objects and positions significantly. Transformer in Transformer (TNT) [43] not only considers self-attention between patches but also within patches. Therefore, the authors embedded an inner Transformer layer in the outer Transformer to enable the network to obtain more granular information while obtaining contextual information. In general, three reasons can explain why models based on the Transformer structure can make outstanding contributions to the visual domain. First, an adaptive input design allows the model to encode the input sequence adaptively, making the model adaptable to different lengths of input and, thus, improving its robustness and generalization ability. Second, the self-attention mechanism strengthens the second-order spatial interaction. Third, the use of large convolution kernels can capture global contextual information, which is beneficial to address long-distance dependencies. However, models based on the Transformer architecture have a high CC, so achieving efficient object detection in practice still poses a challenge.

C. Hybrid Models

In order to better combine the advantages of CNNs and Transformer architecture, some researchers have proposed hybrid object detection models based on Transformers and CNNs. For example, an end-to-end object detection model called detection Transformer (DETR) [44] uses Transformers to directly predict object boxes and categories without traditional anchor boxes and nonmaximum suppression (NMS) processes. However, it has slow training speed and requires larger training data, and its detection performance on small objects is poor. Deformable DETR [45] improves the robustness of deformable objects by adding deformable convolution. It exhibits better detection performance on multiscale and partially occluded objects, but it has high CC and long training time. ACmix [46] combines the advantages of self-attention and convolution and reduces CC compared with self-attention modules, but the CC of channel square cannot be disregarded.

In general, the self-attention mechanism of Transformer can effectively process information of different scales and length ranges, thus avoiding the need for multilayer convolutional networks in traditional object detection methods. Although Transformer-based models, such as DETR, have achieved or even exceeded the performance of some traditional CNN-based object detection models in object detection tasks, their CC explosively increases with an increase in input sequence length, limiting their practical applications.

III. Method

Starting from the successful factors of self-attention in the Transformer architecture, we consider the impact of CC, higher order spatial and channel interactions, and sufficient receptive fields on the network. To address these considerations, we propose a module called CS^n, which breaks free from the CC of $O(N^2)$ while maintaining a sufficient receptive field. The CS^n module can be easily integrated into the existing neural network architectures, replacing conventional modules.

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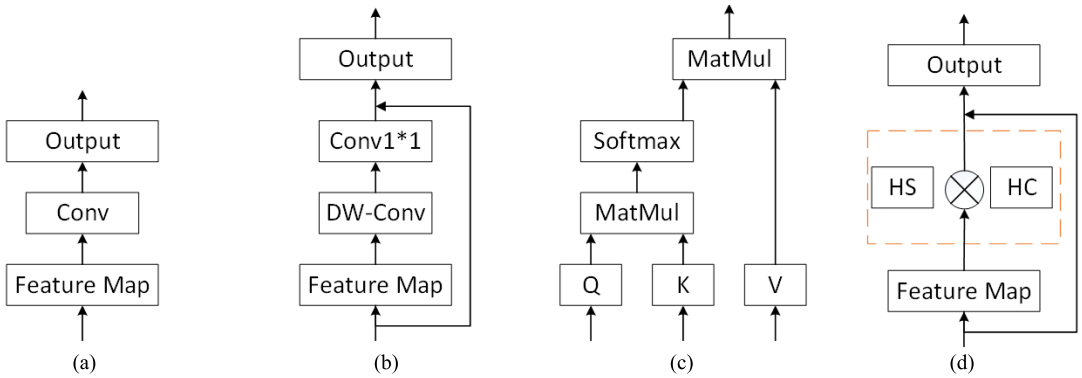

Fig. 1. CS^n and multiple types of convolution structures. (a) Normal convolution structure. (b) Depthwise separable convolution structure with residual connection. (c) Self-attention structure. (d) CS^n structure, and the detailed structure of the orange dashed part is shown in Fig. 6.

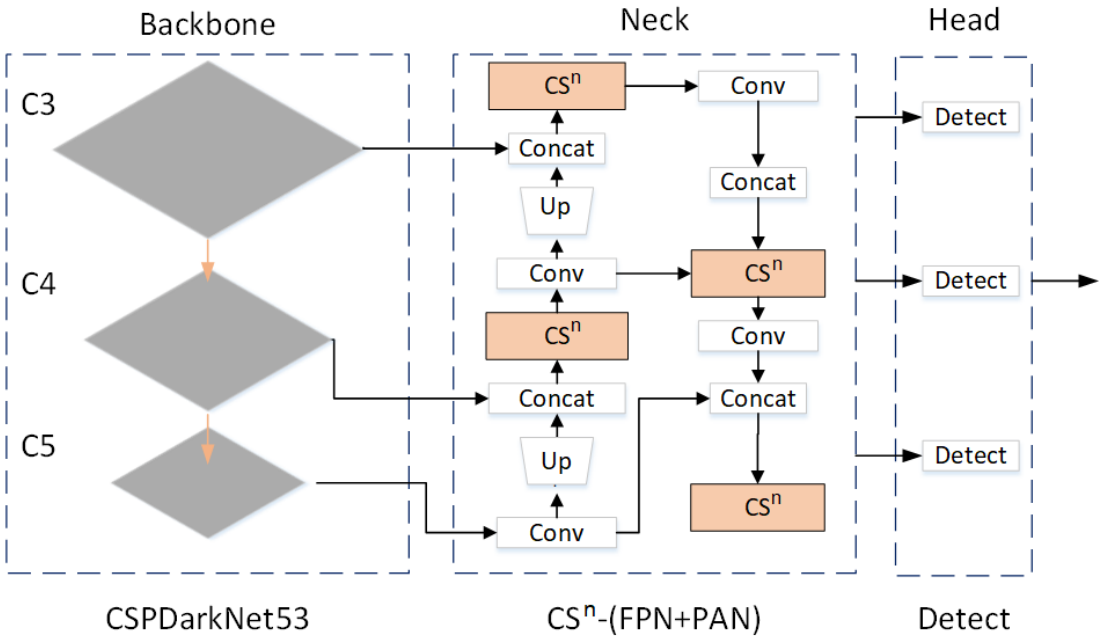

Fig. 2. Network architecture diagram. The backbone of the network adopts CSPDarkNet-53, the neck utilizes CS^n-(FPN + PAN), and the head consists of a generic detection component.

The CS^n module has three major advantages. First, it incorporates large convolutional kernels [27] (e.g., 7 × 7, 13 × 13, or even larger), allowing for the adjustment of kernel size based on different tasks through kernel control switches. This feature facilitates the capture of long-range dependency. Second, it enables arbitrary higher order spatial and channel interactions. While the success of self-attention in the visual domain primarily lies in its second-order spatial interaction [24], it overlooks higher order channel interactions and higher order spatial interactions. The CS^n module, designed in this article, takes these factors into account. Third, it enhances feature extraction capability without significantly increasing the number of parameters. The structure of the CS^n module, along with other convolutional modules, is illustrated in Fig. 1. On the basis of CS^n, we propose a novel and versatile neck structure, CS^n-(FPN + PAN), which is specifically designed for small object detection in remote sensing applications.

The overall architecture of the network is depicted in Fig. 2. Through extensive experiments, we demonstrate that incorporating the proposed CS^n module into the baseline model for object detection outperforms using a single self-attention structure. This enhancement enables the network to improve its ability to recognize small targets in remote sensing, such as maritime vessels, while simultaneously reducing the chances of false positives and false negatives for small object detection, without significantly increasing CC. In addition, we propose a parameter lightweight solution specifically designed for large convolutional kernels within the CS^n module. The specific details of this solution are illustrated in Fig. 3, which enhances the network's global modeling capability while maintaining accuracy.

A. Larger Convolutional Kernels

First, in the Transformer model, the self-attention mechanism often utilizes large convolutional kernels, such as 7 × 7, 13 × 13, or even 25 × 25, which allows for contextual modeling across the entire input sequence and enables strong modeling capability of contextual information. This is beneficial to extract more global features and improve the model's performance. Second, the self-attention mechanism can be computed in parallel, resulting in higher computational efficiency when using large convolutional kernels. By contrast,

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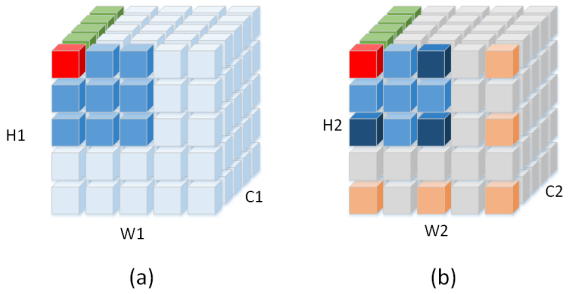

Fig. 3. Segmentation methods for large convolutional kernels. (a) Segmentation method used by MobileNet [47], which decomposes the kernel into two parts. The first part uses a $k \times k$ depthwise separable convolution on $H1$ and $W1$, while the second part uses a 1×1 convolution on $C1$. (b) Segmentation method used in this article, which is divided into three parts. The first part uses a $k \times k$ grouped convolution ($g = C/2^{(O-1)+1}$), the second part uses a grouped dilated convolution ($g = C/2^{(O-1)+1}$), and the third part uses a 1×1 convolution.

the adoption of large convolutional kernels in CNN models leads to an exponential increase in the number of parameters, which can have negative effects. In CNN models, however, convolution operations are typically performed sequentially, and thus, the computational efficiency of the model may be affected when using large convolutional kernels.

To combine the advantages of large convolutional kernels in the Transformer and achieve better contextual modeling capability, we draw inspiration from MobileNet [47] and LAN [48] and propose a segmentation approach for large convolutional kernels in neural networks to reduce the number of parameters. However, these two methods are unsuitable for capturing information interaction in higher order processes. Huang et al. [49] inferred that the use of grouped convolutions in networks with grouped convolutions can leverage the sparsity of diagonal structures to learn information across network dimensions in a more structured manner and facilitate the learning of highly correlated filter features [50], [51]. This approach not only significantly reduces the number of parameters in the network but also prevents overfitting, similar to a regularization effect, enabling the optimizer to learn more accurate and efficient deep networks.

To apply our segmentation method to networks with higher order interactions, we devise rules for the high-order information interaction process. First, we divide the output features into multiple blocks along the higher order spatial and channel dimensions, as described in (1), n represents the order, while $n - O - 1$ is used to arrange the dimensions in an ascending order; this approach allows us to capture a greater amount of information in the higher order dimensions, facilitating the extraction of valuable higher order information. In order to strike a balance between CC and accuracy of recognition, we design grouping formulas that are applicable to arbitrary orders through multiple comparative experiments, as shown in (2). Finally, the visualization of our segmentation method is depicted in Fig. 3. We achieve the segmentation of large convolutional kernels in the convolutional network using the formula shown in (3)

$$C_{\text{out}} = \frac{C}{2^{n-O-1}}, \quad 0 \le O \le n-1 \tag{1}$$

$$g = \frac{C_{\text{in}}}{2^{(O-1)+1}} \quad , 0 \le O \le n-1 \tag{2}$$

$$\text{Conv}_{\text{large}_k} = \text{Conv}_L\left(\text{Conv}_{\text{D},g=\frac{C}{2^{O-1}}}\left(\text{Conv}_{\text{DW},g=\frac{C}{2^{O-1}}}(X)\right)\right). \tag{3}$$

In the formula, X represents the input feature vector, Conv_{DW} and Conv_{D}, respectively, refer to depthwise separable convolution and dilated convolution, and $g = C/2^{(O-1)+1}$ indicates that the input feature is divided into g groups.

In our segmentation method, the kernel size of depthwise separable convolution and dilated convolution is calculated using (4). For convolutions with a kernel size of 23×23 or even larger [27], the kernel size of K_{DW} can be manually reduced on the basis of the input and output feature map sizes by using (5)

$$K_{\text{D}} = \left\lceil \frac{K}{d} \right\rceil, \quad K_{\text{DW}} = \left\lfloor \frac{K+d-1}{d} \right\rfloor \tag{4}$$

$$\text{Out} = \left\lceil \frac{W - K_{\text{D}} + 2P}{S} \right\rceil + 1 \tag{5}$$

where W and Out represent the size of the input and output feature maps, respectively, and S and P represent the stride and padding operations, respectively.

B. Arbitrary Higher Order Spatial and Channel Interactions

In feature extraction, high-order spatial and channel interactions in neural networks consider not only spatial features but also the interaction between channels. This procedure can improve feature discriminability and result in richer feature representations. Self-attention achieves second-order spatial interactions by computing the similarity between each position and all other positions, followed by weighted summation to obtain a representation for each position. This step allows for interaction between all positions in a sequence, but leads to a spatial complexity of $O(N^2)$. However, the spatial complexity of neural networks is the first order, as it only depends on the network structure and the number of parameters, and is independent of the input data size. This natural limitation of neural networks results in a loss of high-order spatial and channel modeling capabilities. From the perspective of convolution, normal convolutional layers have large parameter sizes, and recursive convolutional operations can lead to significant computational cost, as shown in Fig. 4(a). To reduce computational cost, many models use depthwise separable convolution, as shown in Fig. 4(b). This module independently convolves each channel in the input layer and does not effectively utilize the feature information of different channels at the same spatial position.

To overcome this limitation, we use neural networks to simulate self-attention structures. In extracting spatial and channel features, we propose a segmentation method for large convolutional kernels and use group convolution to reduce the model parameter size while fully extracting effective features. The network structure of our approach for segmenting large convolutional kernels and extracting features is shown in Fig. 5. Finally, we introduce arbitrary high-order spatial and channel interactions by using cascading recursive convolution and construct the CS^n module, which outperforms the self-attention module of Transformer. The network structure of the CS^n module is shown in Fig. 6.

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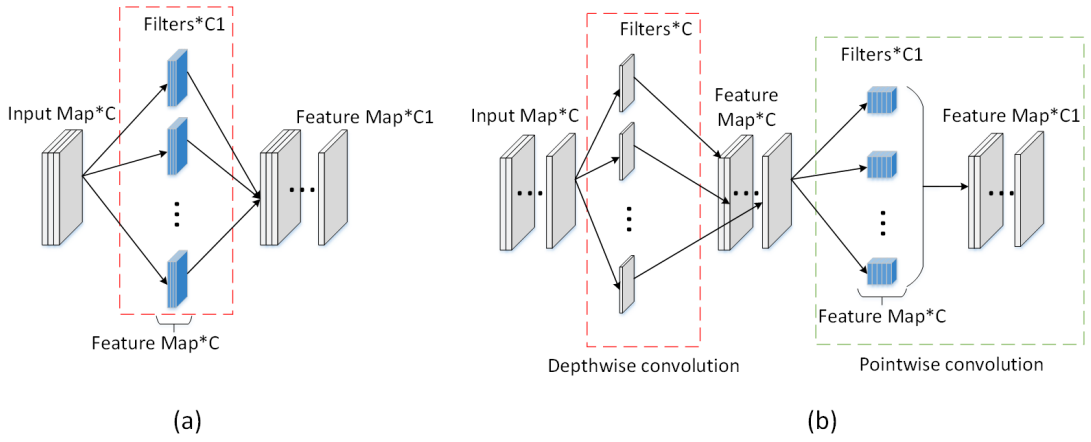

Fig. 4. Feature extraction structures. (a) Normal convolution. (b) Depthwise separable convolution. The red dashed box represents the convolution kernel and its dimension, while the green dashed box represents the pointwise convolution.

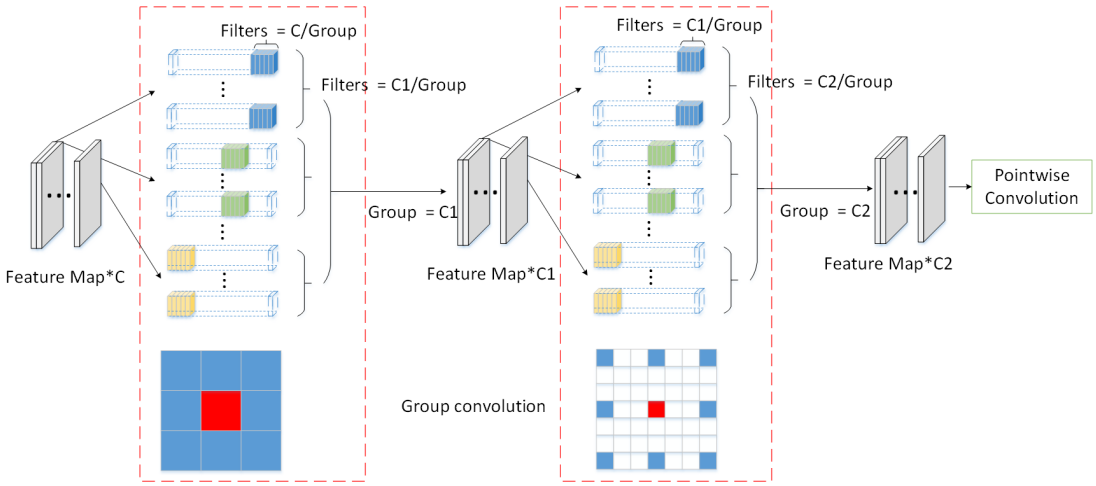

Fig. 5. Feature extraction structure proposed in this article. The first red box represents the use of segmented depthwise separable convolution kernels for feature extraction, and the second red box represents the use of segmented dilated convolution kernels for feature extraction. The green box represents pointwise convolution.

TABLE I
COMPARATIVE ANALYSIS OF CC AND SC AMONG DIFFERENT MODULES

Layer Type	CC	Simplification of CC	SC	Simplification of SC
Self-attention	$O(N^2d)$ [52]	$O(N^2)$	$O(N^2+d^2)$	$O(N^2)$
Covolutional	$O(kNd^2)$ [52]	$O(N)$	$O(k^2d^2)$	$O(1)$
CS^n	$O(Equation\ 14)$	$O(N)$	$O(Equation\ 14)$	$O(1)$

C. Lower Parameter and Computation Costs

The second-order spatial interaction of self-attention is very helpful for feature extraction, but its parameters and CC are proportional to the square of the sequence length and input feature dimension. Therefore, the resulting parameter and CC will be extremely high. Constructing modules for second-order or even higher order spatial and channel interactions is difficult through self-attention and will generate huge computational requirements that most GPU computing power cannot meet. To better demonstrate the performance of the CS^n module, we compare it with self-attention and convolution in terms of CC and space complexity (SC). The comparative results are shown in Table I. Here, CC and SC, respectively, refer to the computational workload and parameter quantity generated by the module. In this context, N represents the sequence length (in the case of images, it denotes the product of the feature map's width and height), and d represents the vector dimension. The input and output channels are both size d.

In accordance with (12), the CC of self-attention can be divided into two steps. First, matrix multiplication $Q_{N\times d}Q_{N\times d}^T$ is performed, resulting in matrix $E_{N\times N}$. This step has a CC of $O(N^2d)$. Then, matrix E is multiplied by V, i.e., $E_{N\times N}V_{N\times d}$, yielding matrix $F_{N\times d}$. This step also has a CC of $O(N^2d)$. Therefore, after disregarding the constants, the final CC of self-attention is $O(N^2)$. Meanwhile, the CC of convolution in image processing is calculated using (13). Here, d_{in} and d_{out} represent the input and output dimensions, respectively, and k denotes the kernel size. The product of the feature map's width and height is denoted as N. Hence, the CC of convolution, disregarding the constant terms, is $O(N)$. If self-attention is recursively applied in a hierarchical manner to obtain interaction information in higher orders, there will be redundant computations in the first step. As a result, the

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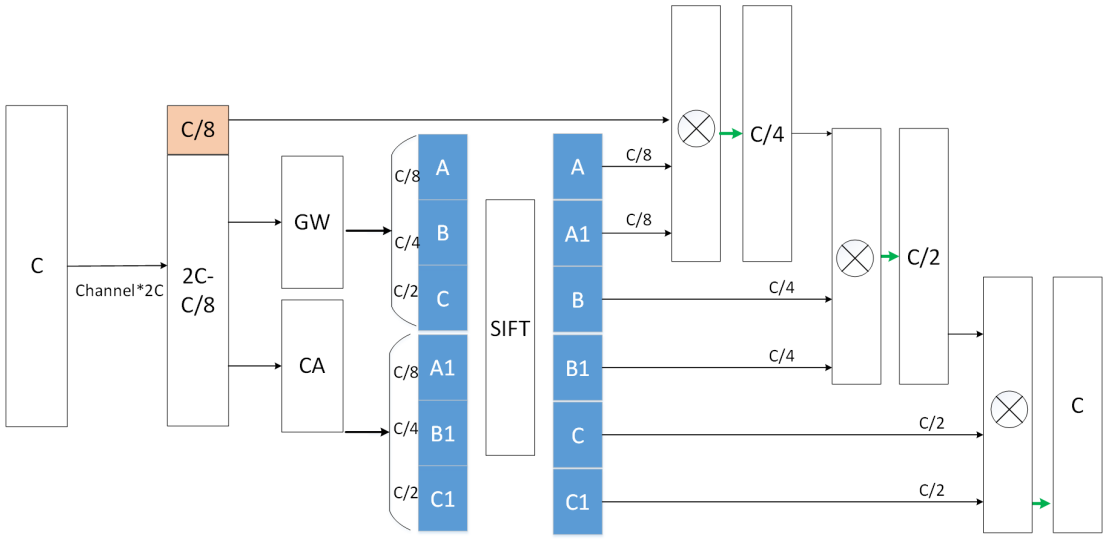

Fig. 6. Network structure of the CS^n module. Same important feature transform (SIFT) represents the reordering of the blockwise features, and the green arrows represent the convolutional operations. The numbers inside the orange box indicate the number of groups. Among them, $O = 4$.

CC in higher orders increases exponentially. To address this issue, we have employed the approach of cascaded recursive convolution to avoid the explosive growth of computational parameters.

In accordance with Fig. 6, the CC of the proposed CS^n module in this article consists of three main components: they are FLOPs(GW), FLOPs(CA), and FLOPs(RE), where RE refers to the recursive convolution operation. Their respective formulas are as follows:

$$\begin{aligned}\text{FLOPs(GW)} &= \text{HW}\sum_{i=1}^{3}k_i^2\sum_{O=0}^{n-1}\frac{d/g}{2^{n-1-O}}\\ &= 2\text{HW}d/g\left(1-\frac{1}{2^n}\right)\sum_{i=1}^{3}k_i^2. \end{aligned} \tag{6}$$

In (6), $\text{HW}\sum_{i=1}^{3}k_i^2\sum_{O=0}^{n-1}(d/g/2^{n-1-O})$ is obtained based on the proposed method of large kernel decomposition, as shown in Fig. 3. The latter term in the equation involves the summation of feature map dimensions based on the order

$$\text{FLOPs(CA)} = 2\left(\frac{d}{g}\right)^2. \tag{7}$$

FLOPs(CA) is used to obtain weight information and consists of two linear layers. Therefore, in accordance with (7), the CC of CA can be calculated as $2(\frac{d}{g})^2$

$$\begin{aligned}&\text{FLOPs(RE)}\\ &= 2\text{HW}\sum_{O=0}^{n-2}\frac{d/g}{2^{n-1-o}}\frac{d/g}{2^{n-2-o}} + 2\text{HW}\sum_{O=0}^{n-1}\frac{d/g}{2^{n-1-o}}\\ &= \frac{4}{3}\text{HW}\left(\frac{d}{g}\right)^2\left(1-\frac{1}{4^{n-1}}\right) + 4\text{HW}d/g\left(1-\frac{1}{2^n}\right)\\ &= \text{HW}d/g\left[\frac{4d}{3g}\left(1-\frac{1}{4^{n-1}}\right)+4-\frac{1}{2^{n-2}}\right]. \end{aligned} \tag{8}$$

The derivation of the CC for recursive convolution is shown in (8). The equation consists of two steps: recursion and multiplication, and the total CC is the sum of these two. The overall CC of CS^n is expressed in (9). By multiplying the width and height of the feature map, we can convert it into a sequence length N, which can be expressed as (10)

$$\begin{aligned}\text{Total} &= 2\text{HW}d/g\left(1-\frac{1}{2^n}\right)\sum_{i=1}^{3}k_i^2 + 2\left(\frac{d}{g}\right)^2\\ &\quad + \text{HW}d/g\left[\frac{4d}{3g}\left(1-\frac{1}{4^{n-1}}\right)+4-\frac{1}{2^{n-2}}\right]. \end{aligned} \tag{9}$$

When the order is very large, simplifying (10) yields (11). It can be observed that the CC of Total(N) has a maximum limit, as evident from the sequence N. This condition implies that the CC remains at the first-order level even as the order increases. Thus, a conclusion can be drawn that the CC of the CS^n module does not exceed $O(N)$, preventing the model from experiencing explosive growth in computational requirements.

We conduct a simple comparison of the spatial complexity between self-attention and convolution (considering only the major parameter layers). The spatial complexity of self-attention is $O(4d^2 + N^2H)$, where H represents the number of attention module heads. The first part of the polynomial represents the parameters generated by the Q, K, and V matrices, and the output matrix, while the second part represents the parameters generated by the attention weights. Simplifying this expression yields a spatial complexity of $O(N^2)$ for self-attention. Meanwhile, in accordance with (14), it can be observed that the parameter count of convolution is independent of the feature map size [41], which means it is independent of the sequence N. Therefore, the spatial complexity of convolution is $O(1)$. Since the CS^n module incorporates recursive convolution, the spatial complexity remains independent of the sequence N. As a result, the CS^n module achieves high-order interaction while maintaining a controlled CC

$$\begin{aligned}&\text{Total}(N)\\ &= 2Nd\left(1-\frac{1}{2^n}\right)\sum_{i=1}^{3}k_i^2 + 2\left(\frac{d}{g}\right)^2\end{aligned}$$

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TABLE II
COMPARISON OF DIFFERENT FEATURE EXTRACTION MODULES IN TERMS OF FUNCTIONALITY

Attribute	S-Conv [14]	Self-A [24]	LKA [48]	g^nconv [37]	CS^n
Local receptive field	✓	✓	✓	✓	✓
Long distance dependency	×	✓	✓	✓	✓
Spatial adaptive	×	✓	✓	✓	✓
Channel adaptive	×	×	✓	✓	✓
High-Order spatial interaction	×	✓	×	✓	✓
High-Order channel interaction	×	×	×	×	✓
Computational complexity	O(n)	O(n^2)	O(n)	O(n)	O(n)

TABLE III
COMPARISON OF PARAMETER COUNTS FOR DIFFERENT FEATURE EXTRACTION MODULES UNDER DIFFERENT FEATURE MAP DIMENSIONS

	Standard type	Decomposition type	Decomposition type	Rate1	Rate2
Cin \ Cout	s-conv	Mobilenet [47]	CS^n	S/ours	M/ours
Cin =32\ Cout =32	173,056	6,432	4,224	41.0	1.5
Cin =64\ Cout =64	692,224	14,912	10,496	66.0	1.4
Cin =128\ Cout =128	2,768,896	38,016	29,184	94.9	1.3
Cin =256\ Cout =256	11,075,584	108,800	91,136	121.5	1.2
Cin =512\ Cout =512	44,302,336	348,672	313,344	141.4	1.1

TABLE IV
EXPERIMENTAL ENVIRONMENT CONFIGURATION

Hardware and software configuration	Environment setup
Operating System	Ubuntu 20.04
CPU	Xeon(R) Silver 4210R CPU
GPU	[GeForce RTX 2080 Ti] * 4
memory	64GB
Python version	3.8
CUDA version	11.3

$$+ Nd/g\left[\frac{4d}{3g}\left(1-\frac{1}{4^{n-1}}\right)+4-\frac{1}{2^{n-2}}\right] \tag{10}$$

$$\text{FLOPs}\left(g^n\text{conv}\right) \leq O\left(2Nd/g\sum_{i=1}^{3}k_i^2+2\left(\frac{d}{g}\right)^2+Nd/g\left[\frac{4d}{3g}+4\right]\right) \leq O(N) \tag{11}$$

$$\text{Attention}(Q,K,V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V \tag{12}$$

$$\text{CC} = d_{\text{in}}\,K^2Nd_{\text{out}}\,. \tag{13}$$

To further demonstrate the advantages of the CS^n module, we introduce it from three aspects: performances, amount of parameters, and process of derivation. First, we compare the CS^n module with normal convolution, self-attention mechanism, large kernal attention (LKA), and g^nconv in terms of functionality, and the comparison results are shown in Tables II and III. It can be seen that our CS^n module has all the comparative advantages.

Second, we compare the parameter counts generated by different feature extraction modules. We set the input and output feature dimensions to 32, 64, 128, 256, and 512 and then calculated the parameter counts for both standard type and decomposition type. The results are shown in Table III. Notably, the convolution kernel segmentation method used in MobileNet [47] is shown in Fig. 3(a), while we used in this article is shown in Fig. 3(b) The calculation method for the amount of parameters in this article is shown in (14), where K is the size of the convolution kernel. In Table III, the default kernel size is set as $K = 13$, and C_{in} and C_{out} represent the dimensions of the input and output features. In the same environment, Rate1 represents the ratio of parameters in a single standard convolution to parameters in a single CS^n. Rate2 represents the ratio of parameters in a single convolution in MobileNet [47] to parameters in a single CS^n. The numerical value indicates that the amount of parameters of the former is several times that of the latter. g is the number of groups in the grouped convolution

$$\text{Para} = K \times K \times \frac{C_{\text{in}}}{g} \times \frac{C_{\text{out}}}{g} \times g. \tag{14}$$

Finally, we use a group-level recursive concatenation method to multiply the parts of the features corresponding to the same order of spatial and channel dimensions with the output features of the previous order, generating higher order feature information. Due to the efficiency of convolution, the parameter count and CC of CS^n module for higher order features are still $O(N)$. The following paragraph explains the specific derivation process of this module.

On the basis of Fig. 6, we first double the dimension of the input features using (15) and then group the features using (16). Equations (17) and (18) generate the first-order interaction spatial and channel features, as shown in the orange part of Fig. 6. Equation (19) reorders the blue part of Fig. 6 to obtain the corresponding spatial and channel information in the feature dimensions. Finally, by utilizing (20), the model interacts the features of each block in the current layer with the features from the previous block through elementwise multiplication, resulting in the final output features. The recursion mentioned here refers to the continuous elementwise multiplication operation depicted in Fig. 6. Through this recursive process, features that appear later in the sequence retain more higher order information. Consequently, the amount

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of feature interaction in higher order dimensions is greater. In the recursive concatenation process, (20) is used to align the feature channels with the output of $\emptyset_{\text{out}}$ () in the feature dimension

$$\emptyset_{\text{in}}(x) \in R^{\text{WH}\cdot C} = X \in R^{\text{WH}\cdot 2C} \tag{15}$$

$$\left[q_0{}^{\text{WH}\cdot C}, k_0{}^{\text{WH}\cdot C}\right] = X \in R^{\text{WH}\cdot 2C} \tag{16}$$

$$q_1 = \left[q_0 \odot k_0\right] \in R^{\text{WH}\cdot C} \tag{17}$$

$$y_{\text{out}} = \emptyset_{\text{out}}(q_1) \in R^{\text{WH}\cdot C} \tag{18}$$

$$\left\{q_0{}^{\text{WH}\cdot C}, \left[k_0{}^{\text{WH}\cdot C} \ldots, k_{O-1}{}^{\text{WH}\cdot C}\right], \left[v_0{}^{\text{WH}\cdot C} \ldots, v_{O-1}^{\text{WH}\cdot C}\right]\right\} = f_{\text{split}}(X) \in R^{\text{WH}\cdot\left(C_0+2\sum_{i=0}^{O-1} C_i\right)} \tag{19}$$

$$q_{o+1} = g_o(q_o) \odot h_o(k_o) \odot l_o(v_o)/\alpha, \quad O = 0, 1 \ldots, n-1 \tag{20}$$

$$g_o = \begin{cases} \text{Identity}, & O = 0 \\ \text{Linear}(C_{O-1}, C_O), & 1 \le O \le n-1. \end{cases} \tag{21}$$

The high-order modulation switch is set by g_o, and $g_o()$ performs group convolution on the historical fused features, while $h_o()$ and $l_o()$ interact with high-order channel and spatial features, respectively. $(1/\alpha)$ is the regulating factor for stable training, and its value is set to 3 in this article.

D. Loss Function

The formula for the overall loss function used in this article is as follows:

$$\text{Total Loss} = \beta_1 \text{FL}(p_t) + \beta_2 L_{\text{CIoU}}\left(b^{\text{loc}}, b^{\text{gt}}\right) + \beta_3 \text{RL}_{\text{CIoU}}\left(b^{\text{refine}}, b^{\text{gt}}\right) \tag{22}$$

where $\text{FL}(p_t)$, $L_{\text{CIoU}}(b^{\text{loc}}, b^{\text{gt}})$, and $L_{\text{CIoU}}(b^{\text{refine}}, b^{\text{gt}})$ denote the classification, localization, and regression losses, respectively. Here, b^{loc}, b^{refine}, and b^{gt} refer to the status of candidate boxes in localization, regression, and annotation, respectively. The hyperparameters β_1, β_2, and β_3 are set to 1, 0.5, and 1.0, respectively, in this article. The detailed formulas for the losses can be found in (23)–(25)

$$\text{FL}(p_t) = -\alpha_t (1-p_t)^{\gamma} \log(p_t). \tag{23}$$

p_t represents the probability value of the cross-entropy loss function, and $-\log(p_t)$ is the cross-entropy loss. α is the weight parameter between classes, and $(1-p_t)^{\gamma}$ is the adjustment factor for easy and difficult samples

$$V = \frac{4}{\pi^2}\left(\arctan\frac{w^{\text{gt}}}{h^{\text{gt}}} - \arctan\frac{w^p}{h^p}\right) \tag{24}$$

$$L_{\text{CIoU}} = \frac{d^2}{L^2} + \frac{V^2}{(1-\text{IoU})+V}. \tag{25}$$

The variables w^{gt} and h^{gt} represent the width and height of the ground-truth (GT) box, while w^p and h^p represent the width and height of the predicted box. L and d denote the Euclidean distance between the centers of the predicted and GT boxes and the diagonal distance of the minimum bounding rectangle, respectively.

IV. Experiment

In this section, we evaluated the proposed method from two aspects: 1) comparing it with advanced detection methods to demonstrate its effectiveness and 2) conducting experiments on different types of remote sensing datasets to demonstrate its robustness.

A. Experimental Environment

The experiment was conducted on the basis of the PyTorch deep learning framework, and both training and testing are performed on a GPU environment. The specific environment configurations are shown in Table IV.

B. Experimental Datasets

We used three publicly available remote sensing datasets: SSDD [53], HRSID [54], and NWPU-10. The dataset properties are shown in Table V. In order to verify that our proposed method is not limited to SAR datasets, we also conducted experiments on the optical remote sensing dataset NWPU-10.

For the sake of fairness, we conducted multiscale analysis on the three datasets and statistically analyzed the aspect ratios of the objects contained in each dataset. As shown in Fig. 7, it can be seen that the objects in the dataset have multiscale characteristics, and training the model on these datasets can improve the model's robustness to multiscale objects.

For the subsequent experiments, we set the initial learning rate to 0.01, momentum to 0.937, and batch size to 16. Training was performed for 100 iterations on the SSDD dataset and 300 iterations on the NWPU-10 and HRSID datasets, respectively, and the experimental results were compared.

C. Evaluation Metrics

This article uses AP, mAP, Paras, GFLOPs, and FPS as evaluation metrics to select the optimal model for maritime remote sensing targets. The meanings of each evaluation metric are shown in Nomenclature.

The formulas for calculating some of the indicators are as follows:

$$P_d(\text{Precision}) = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{26}$$

$$R_d(\text{Recall}) = \frac{\text{TP}}{\text{TP}+\text{FN}}. \tag{27}$$

Here, d represents the intersection over union (IoU) threshold used to determine whether the detection results are true positives or false positives. In addition, R_d represents the recall rate, P_d represents the precision rate, TP represents the number of true positives, FP represents the number of false positives, and FN represents the number of false negatives

$$\text{AP}_d = \int_0^1 P_d(R_d)\, dR_d \tag{28}$$

$$\text{mAP}_d = \frac{1}{N_c}\sum_{i}^{N_c}(\text{AP}_i)_d \tag{29}$$

$$\text{FPS} = \frac{1}{\text{Time}}. \tag{30}$$

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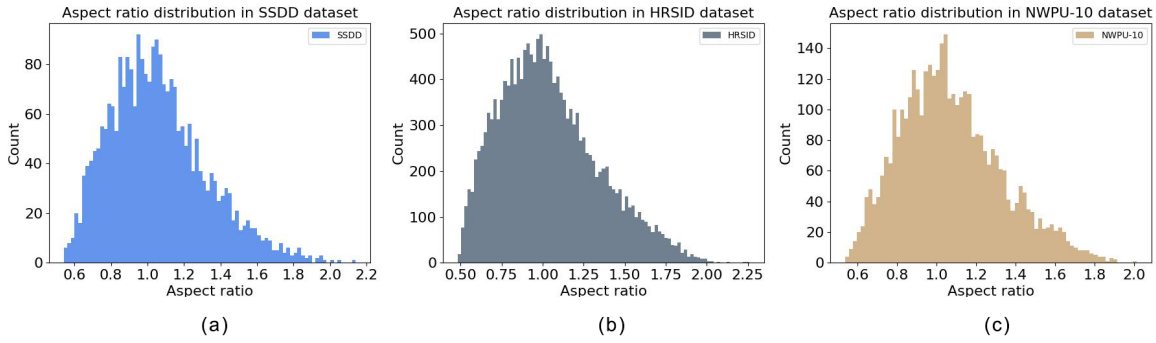

Fig. 7. Distribution of aspect ratios of objects in different datasets, with (a) SSDD dataset, (b) HRSID dataset, and (c) NWPU-10 dataset.

TABLE V
DATASET PROPERTIES TABLE

	Data Type	Source	Amount	Resolution	Polarization mode
SSDD [53]	SAR	RadarSat2 TerraSARX Sentinel1	1160	1m-15m	HH,HV,VV,VH
HRSID [54]	SAR	-	5604	0.5m, 1 m, 3 m	HH,HV,VV,VH
NWPU-10	Optical	Google Earth	800	-	-

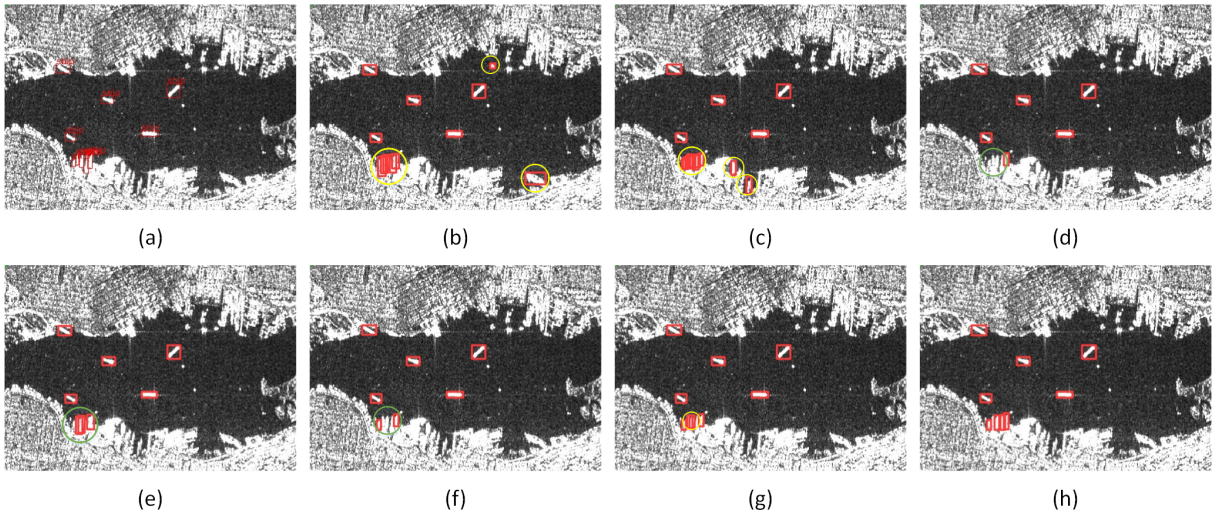

Fig. 8. Detection results of different models on the nearshore data in the SSDD dataset. (a) GT image. (b)–(h) Detection results of Faster region based convolutional neural network with feature pyramid network (R-CNN-FPN), YOLOv3, YOLOv5, YOLOv6, YOLOv7, Improved paddle real-time detection (PRDet), and our CS^{n}Net, respectively. Yellow circles represent false positive detections, and green circles represent missed detections.

TABLE VI
TEST DATA COMPARISON OF DIFFERENT MODELS ON THE SSDD DATASET

Method	**Backbone**	$\mathbf{mAP}_{.5}(\%)$	$\mathbf{mAP}_{.5:.95}(\%)$	**Para(M)**	**FLOPs(G)**
Faster r-cnn-fpn [11]	Resnet50	85.2	41.2	39.8	134.4
YOLOV3 [30]	Darknet-53	88.6	44.3	58.7	155.3
YOLOV5(base) [55]	CSPDarknet53	93.7	62.9	**6.7**	**15.9**
YOLOV6 [32]	RepVGG	96.1	63.9	18.5	45.3
YOLOV7 [33]	CSPDarknet53	94.6	63.3	35.5	105.3
Improved PRDet [56]	Resnet50	96.5	64.3	35.5	132.8
YOLOV5-SwinTransformer [42]	SwinTransformer	94.0	57.9	13.4	105.3
CS^{n}Net	CSPDarknet53	**97.1**	**64.9**	**12.2**	**21.7**

In the preceding equations, N_c represents the number of classes, and AP_i represents the AP for class i. AP is a metric calculated on the basis of the precision–recall curve, and it is used to measure the performance of object detection algorithms. mAP represents the average AP across m classes, and it can evaluate the detection capabilities of the trained model for all classes. The calculation formula is shown in (22). In this article, $mAP_{.5}$ is calculated at an IoU threshold of 0.5, while $mAP_{.5:.95}$ represents the average mAP calculated at different IoU thresholds ranging from 0.5 to 0.95 with a step size of 0.05. The time (s) and FPS for detecting a single optical remote sensing image are defined by (23).

D. Comparison With Baseline and Other Models

The baseline model used in this article is YOLOv5, with the neck component of the network replaced by the proposed

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TABLE VII
TEST RESULTS OF DIFFERENT MODELS ON INSHORE AND OFFSHORE DATA IN THE SSDD DATASET

Method	Backbone	IS $\mathbf{mAP}_{.5}(\%)$	IS $\mathbf{mAP}_{.5:.95}(\%)$	OS $\mathbf{mAP}_{.5}(\%)$	OS $\mathbf{mAP}_{.5:.95}(\%)$
Faster r-cnn-fpn [11]	Resnet50	80.7	42.9	93.7	56.4
YOLOV3 [30]	Darknet-53	90.7	50.8	95.0	51.1
YOLOV5(base) [55]	CSPDarknet53	92.3	51.4	97.7	63.7
YOLOV6 [32]	RepVGG	94.3	52.6	98.4	60.2
YOLOV7 [33]	CSPDarknet53	93.4	52.0	98.1	59.8
Improved PRDet [56]	Resnet50	85.8	44.7	98.5	60.4
YOLOV5-SwinTransformer [42]	SwinTransformer	92.4	50.9	98.0	62.6
CS^{n}Net	CSPDarknet53	**95.1**	**53.1**	**98.8**	**64.6**

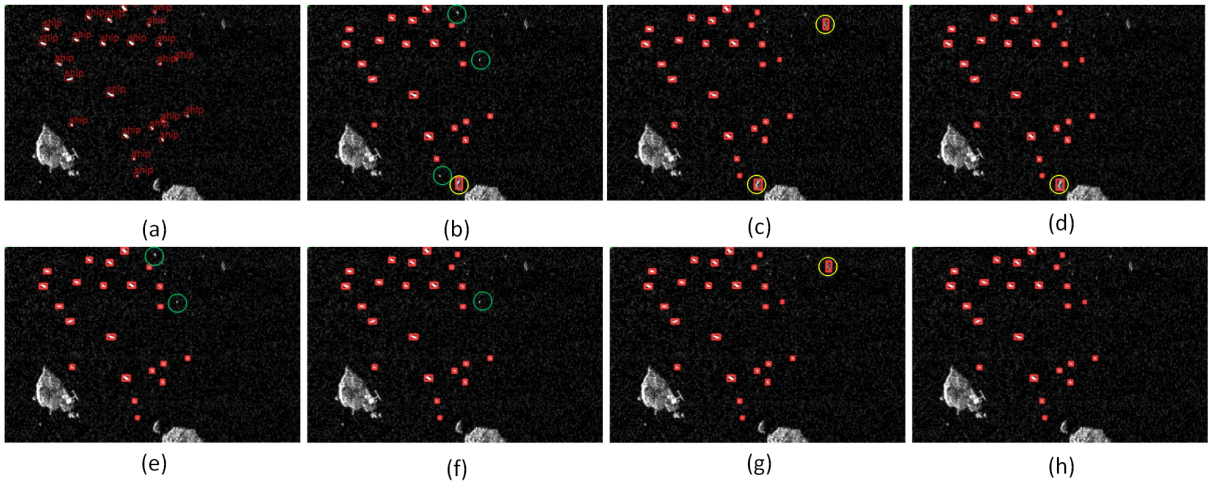

Fig. 9. Detection results of different detection models on offshore data in the SSDD dataset. (a) GT image. (b)–(h) Detection results of Faster R-CNN-FPN, YOLOv3, YOLOv5, YOLOv6, YOLOv7, Improved PRDet, and our CS^{n}Net, respectively. Yellow circles represent false positives, and green circles represent false negatives.

TABLE VIII
TEST RESULTS OF DIFFERENT MODELS ON THE HRDIS DATASET

Method	Backbone	$\mathbf{mAP}_{.5}(\%)$	$\mathbf{mAP}_{.5:.95}(\%)$	Para(M)	FLOPs(G)	FPS
Faster r-cnn-fpn [11]	Resnet50	79.1	46.6	39.82	134.4	16
YOLOV3 [30]	Darknet-53	82.6	52.1	58.7	155.3	30
YOLOV5(base) [55]	CSPDarknet53	86.7	54.4	**6.7**	**15.9**	**49**
YOLOV6 [32]	RepVGG	89.3	59.2	18.5	45.3	28
YOLOV7 [33]	CSPDarknet53	88.7	58.2	35.5	105.3	17
Improved PRDet [56]	Resnet50	90.7	59.8	35.5	132.8	39
YOLOV5-SwinTransformer [42]	SwinTransformer	88.3	54.0	13.4	105.3	19
CS^{n}Net	CSPDarknet53	**91.2**	**60.6**	**12.2**	**21.7**	**45**

TABLE IX
TEST RESULTS OF DIFFERENT MODELS ON THE NWPU-10 DATASET

Method	Backbone	$\mathbf{mAP}_{.5}(\%)$	$\mathbf{mAP}_{.5:.95}(\%)$	Para(M)	FLOPs(G)	FPS
Faster r-cnn-fpn [11]	Resnet50	81.6	45.9	39.8	134.4	7
YOLOV3 [30]	Darknet-53	88.2	51.3	58.7	155.3	30
YOLOV5(base) [55]	CSPDarknet53	90.0	54.8	**6.7**	**15.9**	**49**
YOLOV6 [32]	RepVGG	90.1	55.2	18.5	45.3	45
YOLOV7 [33]	CSPDarknet53	86.9	53.6	35.5	105.3	44
Improved PRDet [56]	Resnet50	89.1	53.3	35.5	132.8	40
YOLOV5-SwinTransformer [42]	SwinTransformer	89.8	53.9	13.4	105.3	22
CS^{n}Net	CSPDarknet53	**90.4**	**55.4**	**12.2**	**21.7**	**48**

CS^n-(FPN + PAN) to construct CS^{n}Net. CS^{n}Net is then compared with state-of-the-art object detection models on the SSDD dataset. The specific experimental results are presented in Table VI.

From Table VI, our proposed model achieved the highest experimental results of 0.971 and 0.649 on $\text{mAP}_{.5}$ and $\text{mAP}_{.5:.95}$, respectively, compared with other advanced object detection models on the SSDD dataset. Although embedding the CS^n module ($O = 5$) in the base model increases the model's parameters and computation, it leads to significant accuracy improvements. To further obtain experimental data for inshore and offshore targets in the dataset, we conducted separate accuracy tests on inshore and offshore data for each model. The test results are presented in Table VII.

The data of inshore (IS) [53] contains a large amount of background information, which can interfere with detection

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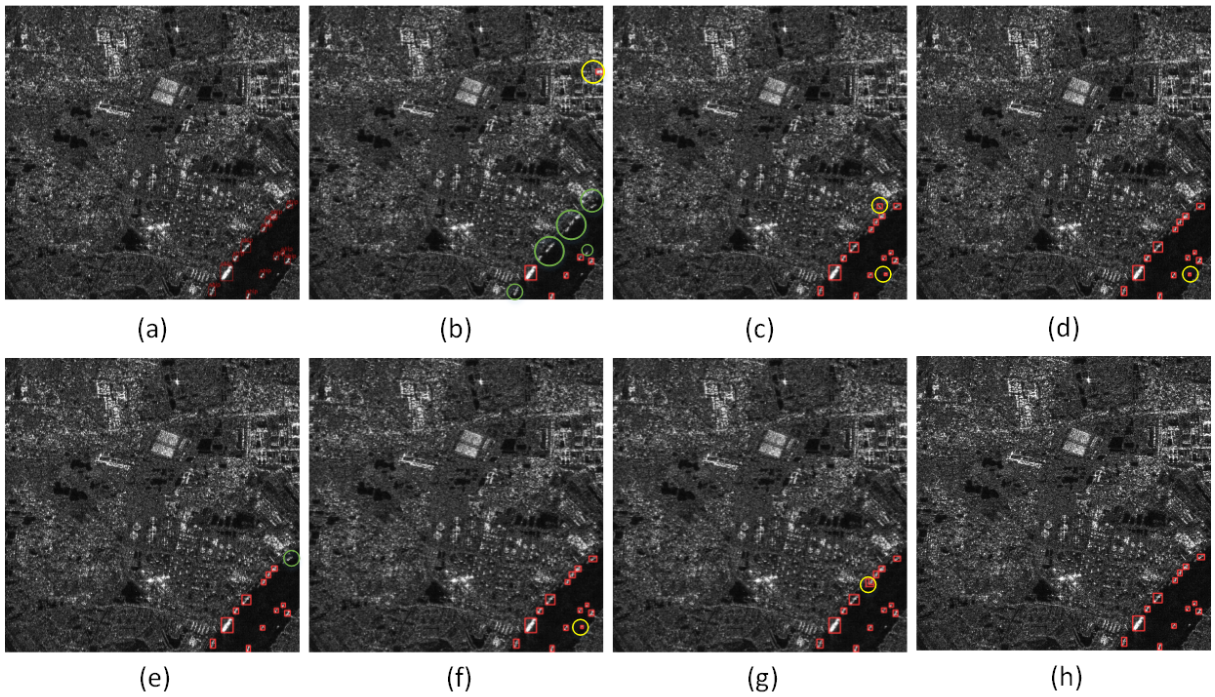

Fig. 10. Detection results of various detection models on nearshore data in the HRSID dataset. (a) GT image. (b)–(h) Detection results of Faster R-CNN-FPN, YOLOv3, YOLOv5, YOLOv6, YOLOv7, Improved PRDet, and our CS^{n}Net, respectively. Yellow circles represent false detection targets, and green circles represent missed detection targets.

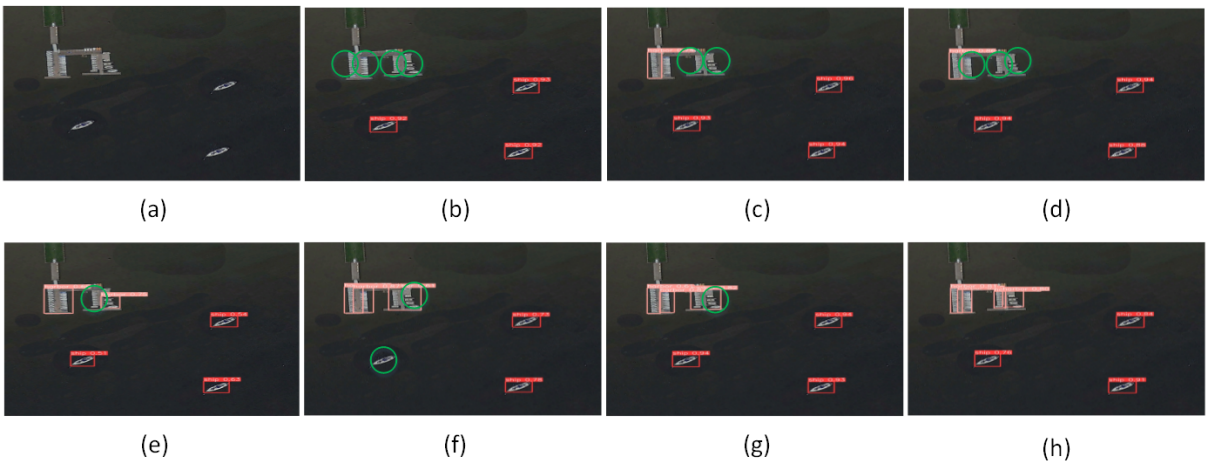

Fig. 11. Detection results of various detection models on the NWPU-10 dataset. (a) GT image. (b)–(h) Detection results of Faster R-CNN-FPN, YOLOv3, YOLOv5, YOLOv6, YOLOv7, Improved PRDet, and our CS^{n}Net, respectively. Yellow circles represent false positives, and green circles represent false negatives.

and easily lead to false detection. The small targets in the offshore (OS) data are densely distributed, which can easily cause missed detection problems. The specific visualization data presented in Figs. 8 and 9. As indicated in Table VII, our model has strong anti-interference ability and remote sensing small target recognition ability, achieving the highest detection accuracy of 0.951 and 0.531, and 0.988 and 0.646 for mAP$_{.5}$ and mAP$_{.5:.95}$ on the inshore and offshore test sets, respectively, which is a 2.8%, 1.7%, 1.1%, and 0.9% improvement over the base model.

Subsequently, we conducted corresponding experimental comparisons on the HRSID dataset [54], and the visual detection results of the near shore are presented in Fig. 10. By observing the test experimental data in Table VIII, our model achieved the highest detection accuracy of 0.912 and 0.606 in mAP$_{.5}$ and mAP$_{.5:.95}$, respectively, which is an improvement of 4.5% and 6.2% compared with the base model. In this experiment, we introduced the FPS indicator. By observing the FPS data in Table VIII, our model achieved a test result of 45, which provides support for embedding the remote sensing detection model into lightweight devices.

To further demonstrate the effectiveness of the CS^n module proposed in this article, we conducted experiments on the optical remote sensing dataset NWPU-10 to compare the detection performance of different models. The test results are presented in Table IX. It can be found that the model with the embedded CS^n module not only has good detection accuracy on the remote sensing SAR dataset, but also has

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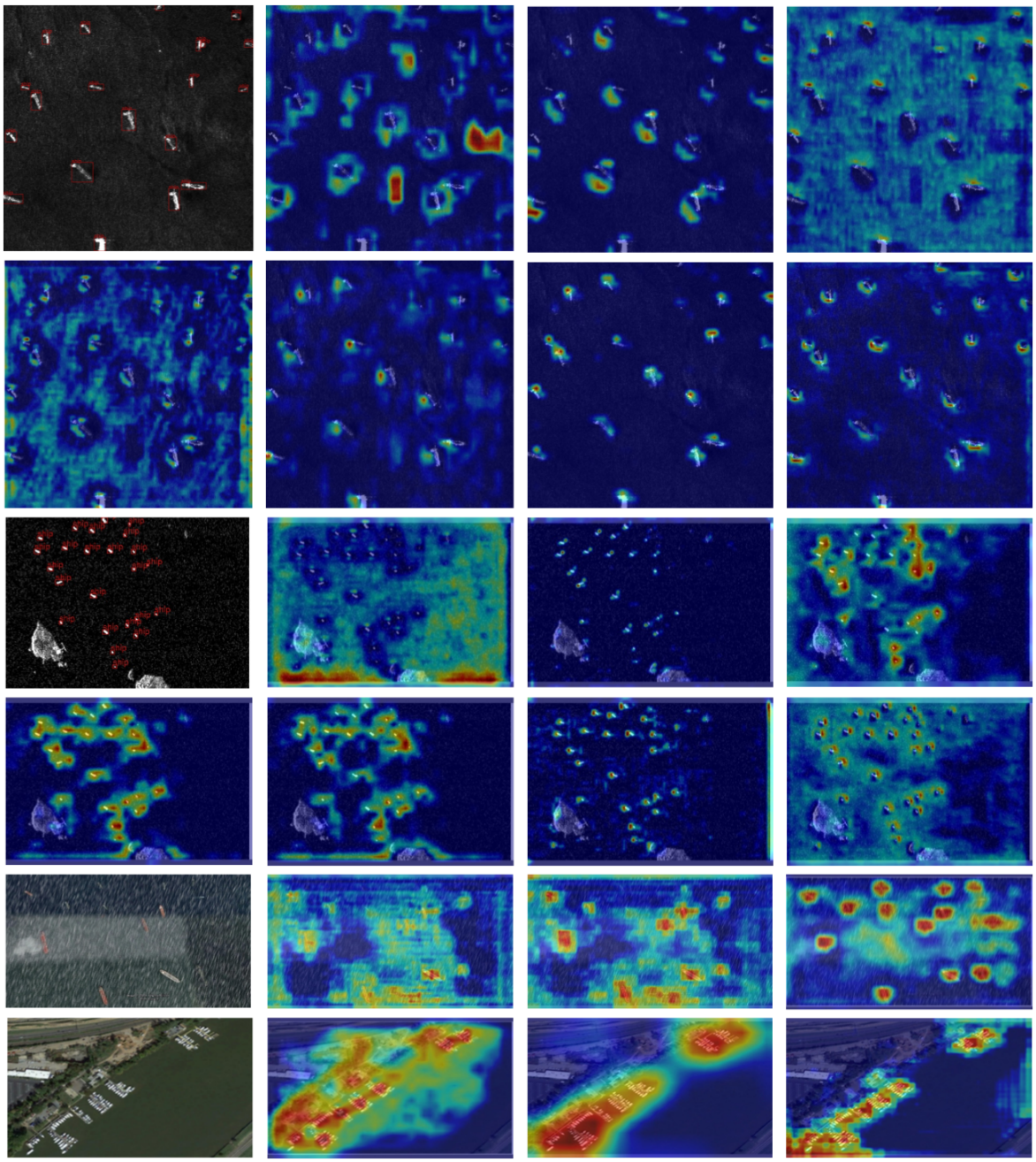

Fig. 12. Heat-map visualization results of various detection models on different datasets. The images are divided into three groups according to the first two rows, middle two rows, and last two rows, respectively, representing the heat-map visualization results of various detection models on the SSDD, HRSID, and NWPU-10 datasets. In each group, from left to right, the original image and the heat map of Faster R-CNN-FPN, YOLOv3, YOLOv5, YOLOv6, YOLOv7, Improved PRDet, and our CS^{n}Net are shown.

efficient detection performance on the optical remote sensing images. Our model achieved the highest precision of 0.904 and 0.554 on this dataset, respectively, which improved by 0.36% and 0.63% compared with the baseline model, and achieved an FPS of 48. The visualization of detection results are shown in Fig. 11.

E. Heat-Map Representation

We visualize the attention level of each network on the targets during the detection process by heat maps. As shown in Fig. 12, the interaction between high-level spatial and channel features can affect the model's performance to different degrees. This condition can enable the model to improve its ability to resist interference from nontargets while focusing on the correct targets to reduce the probability of false alarms and missed detections.

V. Conclusion and Future Work

In this article, we propose a feature aggregation module called CS^n that can be extended to arbitrary higher order spatial and channel interactions to capture long-range dependencies in CNNs similar to the Transformer structure. It inherits the feature extraction capability of large convolutional kernels while reducing the CC by decomposing

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them. Self-attention has shown that the second-order spatial interactions contribute to capturing useful features, but its CC of $O(N^2)$ compared with convolution's $O(N)$ makes it challenging to deploy such networks. Therefore, we introduce arbitrary higher order spatial and channel interactions in neural networks through cascaded recursive convolutions, breaking the limitation of self-attention in only computing second-order spatial interactions and reducing the CC to $O(N)$. Ultimately, we construct CS^{n}Net, a small object detection network for remote sensing, incorporating the CS^n-(FPN + PAN) module.

To address the feature misalignment issue in higher order spatial and channel interactions, we employ group fusion and group convolution methods. Finally, our proposed model outperforms other advanced detection models in remote sensing detection on publicly available SAR datasets, such as SSDD, HRSID, and optical remote sensing dataset NWPU-10. Compared with the baseline model, CS^{n}Net achieves a significant advantage in remote sensing detection, with the improvements of 3.4%, 4.5%, and 0.4% in mAP$_{.5}$ for each dataset, respectively. Importantly, the ideas presented in this article are not limited to remote sensing object detection but can also be applied to other downstream tasks in computer vision.

In our future work, we will delve deeper into exploring useful information from higher order spatial and channel interactions. We aim to construct a high-performance universal feature extraction network that excels in downstream tasks, such as object detection, semantic segmentation, and object classification, while balancing accuracy and computational efficiency.

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